

Experiment 7

Module Name: Robot Frameworks

Experiment Name: Reports and Logs

Problem Statement: Execute a test suite to generate an HTML report containing test details and outcomes, and use the "Log" keyword to write custom log messages for debugging purposes.

Objective: To understand Robot Framework's native output artifacts and enhance traceability by logging dynamic variables during runtime.

Tools, Software, and Concepts Used:

BuiltIn Log keyword, Output files: output.xml, log.html, report.html

Source Code / Implementation Steps:

1. At first create a folder inside that folder create 'Experiment7.robot' and write this

```
*** Settings ***
Library      BuiltIn

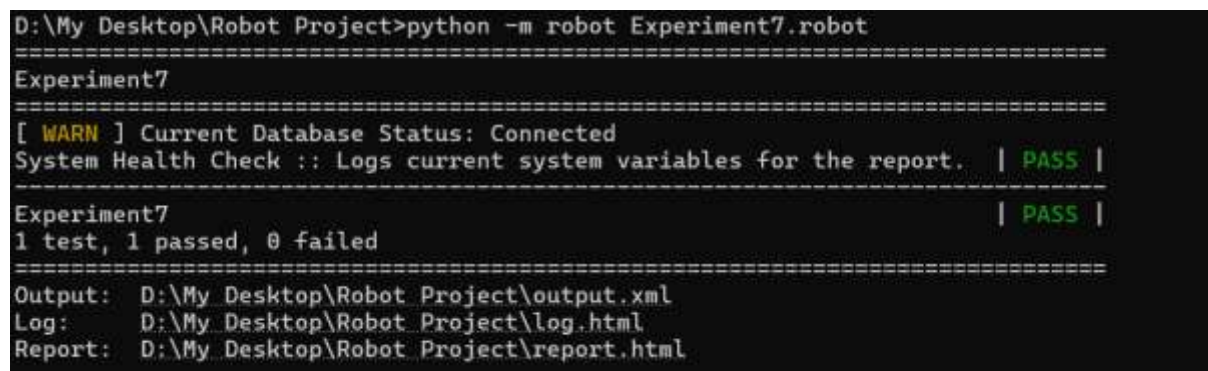
*** Variables ***
${ENVIRONMENT}    Staging
${DB_STATUS}      Connected

*** Test Cases ***
System Health Check
    [Documentation]    Logs current system variables for the report.
    Log      Executing test in the ${ENVIRONMENT} environment.
    Log      Current Database Status: ${DB_STATUS}    level=WARN
    Should Be Equal    ${DB_STATUS}    Connected
```

2. From the folder go to cmd and write this:

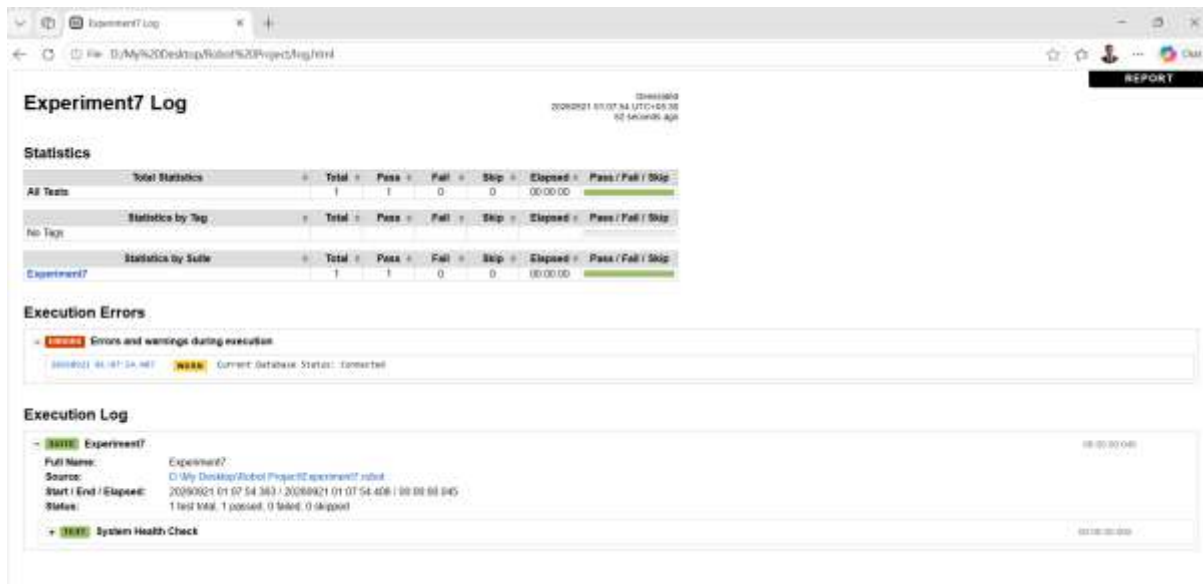
```
>>> python -m robot Experiment7.robot
```

Output with clearly labelled screenshots:

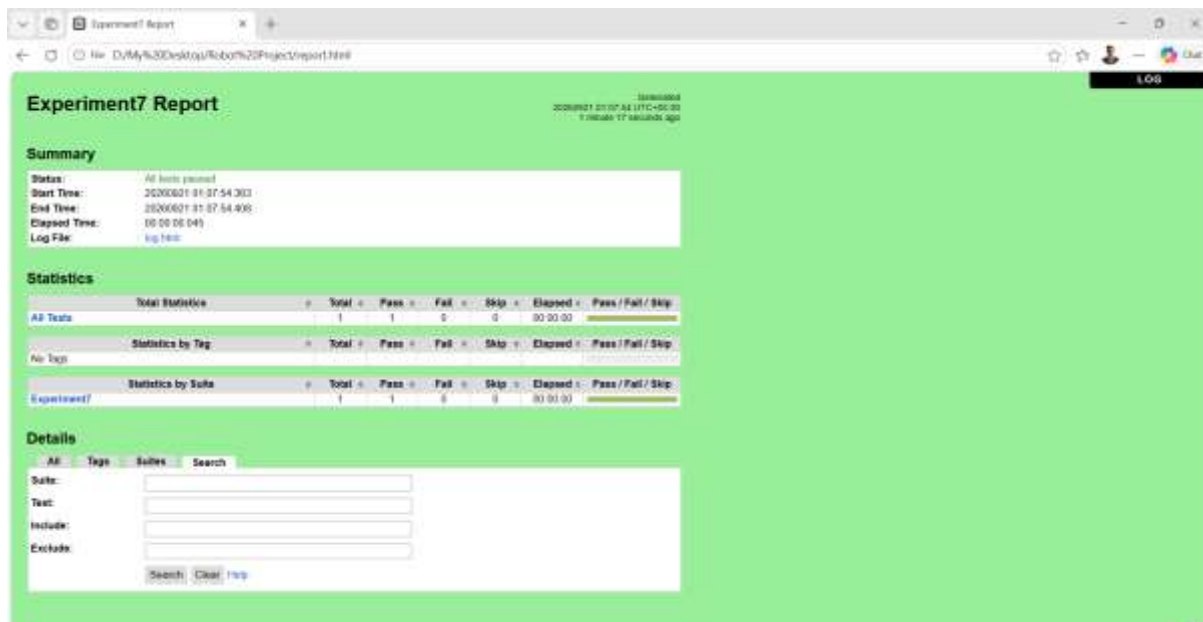


```
D:\My Desktop\Robot Project>python -m robot Experiment7.robot
=====
Experiment7
=====
[ WARN ] Current Database Status: Connected
System Health Check :: Logs current system variables for the report. | PASS |
=====
Experiment7 | PASS |
1 test, 1 passed, 0 failed
=====
Output:  D:\My Desktop\Robot Project\output.xml
Log:     D:\My Desktop\Robot Project\log.html
Report:  D:\My Desktop\Robot Project\report.html
```

Screenshot 1: Terminal Output



Screenshot 2: Output of log.html



Screenshot 3: Output of report.html

Brief Result: Following test execution, Robot Framework automatically generated the XML, Log, and Report artifacts. The custom log messages were accurately captured inside the log.html file.

Observation: The `level=WARN` parameter visually highlights the log entry in yellow inside the HTML file, making it easy to spot during debugging.

Conclusion: Robot Framework's automatic, highly detailed HTML reporting drastically reduces the overhead required to communicate test results to stakeholders.